

## 100/1000Base-T1-TX-E User Manual



| Data       | Version | Description |
|------------|---------|-------------|
| 2024/11/1  | V10     |             |
| 2025/2/1   | V11     |             |
| 2025/11/11 | V12     |             |

## 1 Introduction

### 1.1 Product Series

| Product Name                   | T1 Interface                                                  | Power Supply      | T1 Phy             | Tx Phy    | Size LxHxW(mm) |
|--------------------------------|---------------------------------------------------------------|-------------------|--------------------|-----------|----------------|
| <b>100/1000Base-T1-TX-E</b>    | 15EDG3.81mm-2P<br>MATENET Adapter Board<br>HMTD Adapter Board | Type C<br>DC Jack | Marvell<br>88Q2112 | RTL8211FI | 50 × 20× 83    |
| <b>100/1000Base-T1-TX-TE</b>   | TE MATENET Male                                               | Type C<br>DC Jack | Marvell<br>88Q2112 | RTL8211FI | 50 × 20× 83    |
| <b>100/1000Base-T1-TX-HMTD</b> | HMTD Male                                                     | Type C<br>DC Jack | Marvell<br>88Q2112 | RTL8211FI | 50 × 20× 83    |

### 1.2 Product Description

The BUELEC 100/1000Base-T1-TX-E is a compact, high-performance media converter designed to bridge automotive single-pair Ethernet (100/1000BASE-T1) networks with standard twisted-pair Gigabit Ethernet (100/1000BASE-TX) infrastructure.

Utilizing the Marvell 88Q2112 PHY for T1 interface and Realtek RTL8211FI PHY for TX interface, this converter ensures seamless interoperability between modern automotive Ethernet systems and traditional IT/networking environments.

It is ideal for in-vehicle diagnostics (DoIP), smart cockpit domain controller flashing/upgrades, ADAS camera/radar development and debugging, industrial automation, and fleet management applications.

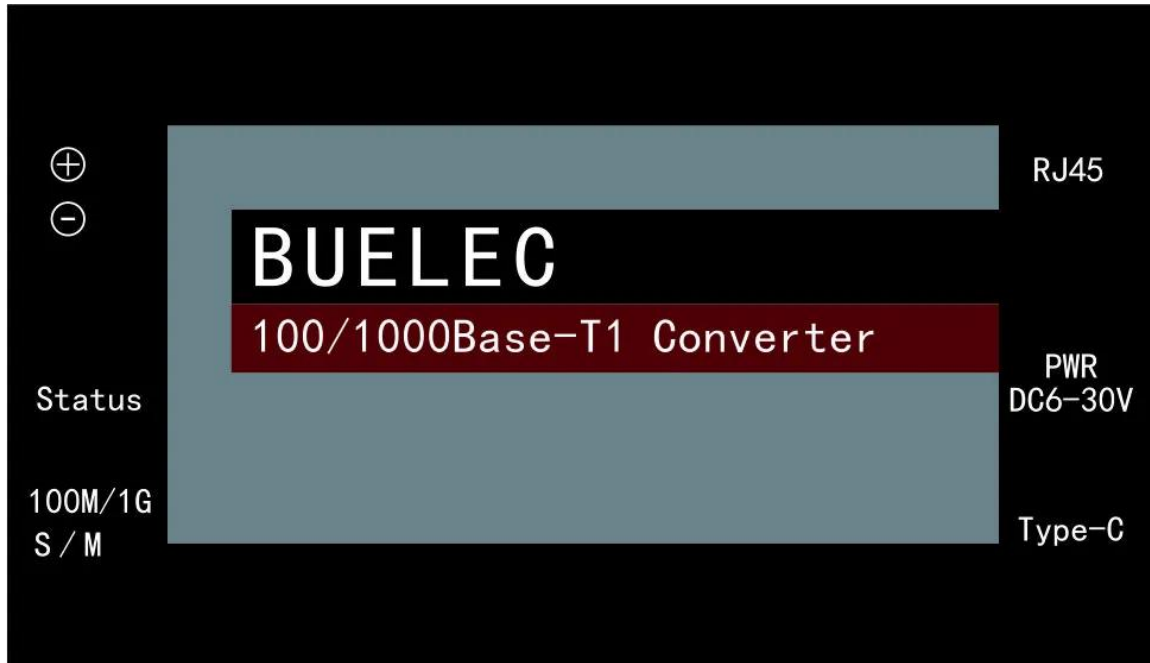
### 1.3 Product Features

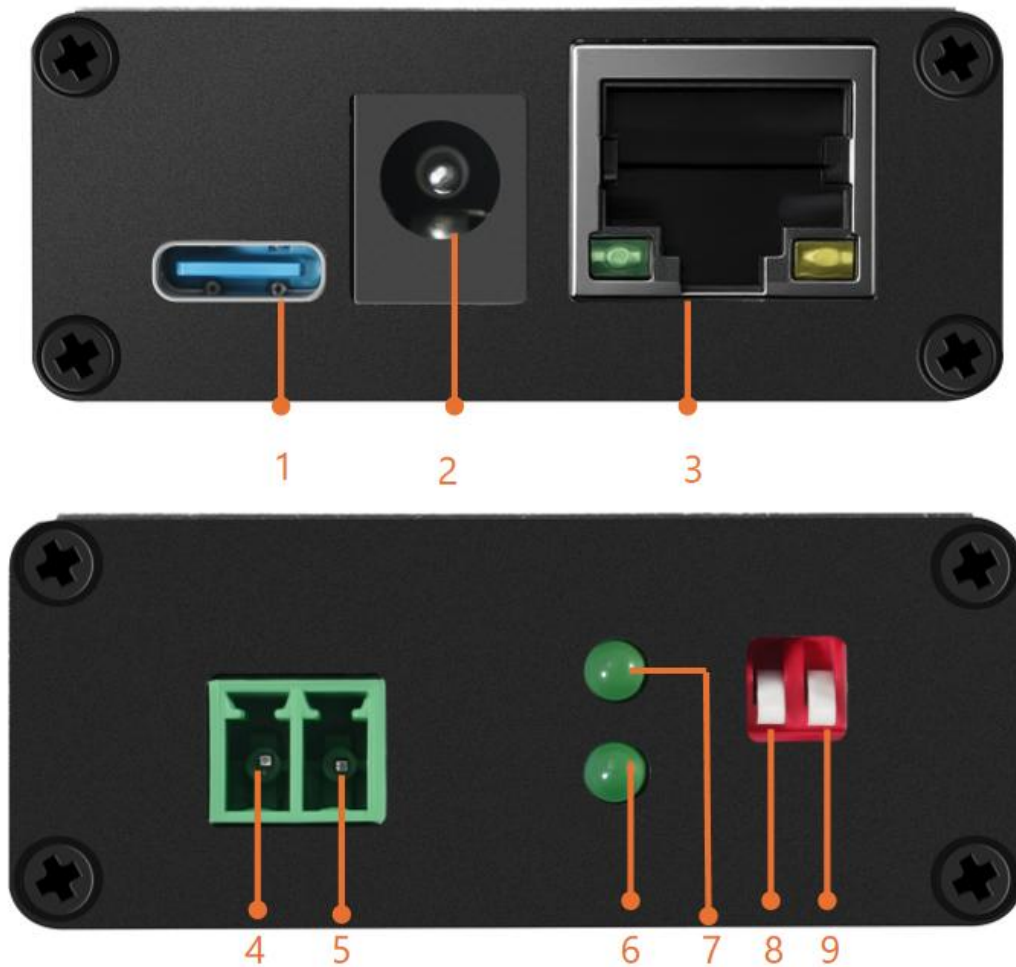
- **Protocol Compliance:** Fully supports IEEE 802.3bw (100BASE-T1) / Fully supports IEEE 802.3bp (1000BASE-T1) for automotive single-pair Ethernet and IEEE 802.3ab (1000BASE-T) for traditional Gigabit Ethernet, enabling reliable data transmission in mixed-network setups.
- **High-Speed Performance:** Delivers up to 1 Gbps full-duplex transmission rates over both T1 (single twisted pair) and TX (RJ45) interfaces, with auto-negotiation for 100/1000 Mbps compatibility and low-latency forwarding.

- **Robust Protection:** Integrated over-voltage, over-current, and ESD (electrostatic discharge) protection safeguards against electrical surges, meeting automotive-grade reliability standards.
- **Industrial-Grade Durability:** Engineered to industrial standards with a wide operating temperature range (-40°C to +85°C), humidity tolerance (0-95% RH non-condensing), vibration resistance, and EMI shielding, making it suitable for harsh environments like vehicles, factories, and outdoor installations.
- **Advanced Networking Support:** Includes IEEE 802.1Q VLAN tagging for traffic segmentation, QoS prioritization, and enhanced security in multi-device ecosystems.
- **Plug-and-Play Design:** Compact form factor (50mm x 20mm x 83mm) with standard T1 differential signal terminal (15EDG3.81mm-2P) on one side and RJ45 port on the other; Power update to support Both Type C and DC Jack(6-30V)

## 2 Hardware

### 2.1 Interfaces and Indicators (Detailed Description)





|     | Interface  | Description                                                                                                                                                                            |
|-----|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (1) | Power      | (DC 5V).                                                                                                                                                                               |
| (2) | Power      | DC Jack (6-30V)                                                                                                                                                                        |
| (3) | RJ45       | 100/1000BASE-T Ethernet port. Connects to PCs, switches, or other network devices using <b>4 standard</b> twisted-pair cable (CAT6 recommended).                                       |
| (4) | T1 +       | 100/1000BASE-T1 differential signal interface using <b>15EDG3.81mm-2P terminal block</b> . Connects to ECU or other automotive Ethernet devices via a single twisted pair.             |
| (5) | T1 -       |                                                                                                                                                                                        |
| (6) | Status LED | Dual-function system and link/activity indicator:<br><ul style="list-style-type: none"> <li>• <b>Solid Green</b>: System self-test passed and device is operating normally.</li> </ul> |
| (7) | Status LED | <ul style="list-style-type: none"> <li>• <b>Solid (T1 Link)</b>: 100/1000BASE-T1 link established.</li> </ul>                                                                          |

|     |                            |                                                                             |
|-----|----------------------------|-----------------------------------------------------------------------------|
|     |                            | • <b>Blinking:</b> Data transmission/reception in progress on T1 interface. |
| (8) | <b>100M/1000M Switch</b>   | UP: 1000M<br>Down:100M                                                      |
| (9) | <b>Slave/Master Switch</b> | UP:Master<br>Down:Slave                                                     |

## 2.2 Typical Applications

PC/ARM ↔ RJ45 ↔ [1000BASE-T1-TX-E] ↔ Automotive Device / Network

| Input Protocol     | Output Protocol | Applications                                                              |
|--------------------|-----------------|---------------------------------------------------------------------------|
| <b>100BASE-Tx</b>  | 100BASE-T1      | Vehicle diagnostics (DoIP), smart cockpit flashing, ADAS sensor debugging |
| <b>1000BASE-Tx</b> | 1000BASE-T1     | High-speed camera/radar data, ECU programming, R&D testing                |

The 1000BASE-T1-TX-E enables seamless conversion between industrial Ethernet and automotive single-pair Ethernet networks.

## 2.3 Product Specifications

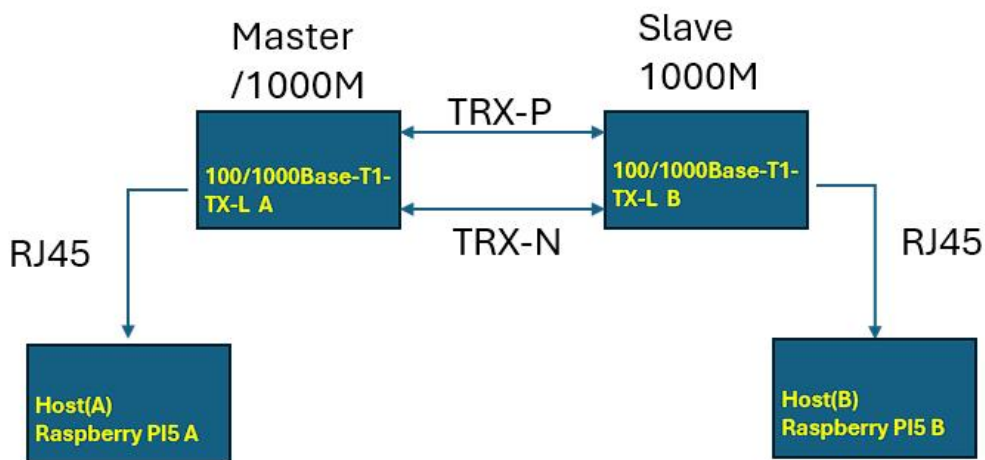
| Parameter                    | Specification             |
|------------------------------|---------------------------|
| <b>Input Voltage</b>         | 5V ± 0.5V                 |
| <b>Operating Current</b>     | ≤ 355mA                   |
| <b>Operating Temperature</b> | -40°C to +85°C            |
| <b>Operating Humidity</b>    | 0~95% RH (non-condensing) |
| <b>Dimensions LxHxW</b>      | 50mm × 20mm × 83mm        |

## 2.4 Automotive Ethernet Transmission Distance

| T1 Speed Mode     | Cable Type                    | Max Distance |
|-------------------|-------------------------------|--------------|
| <b>100BASE-T1</b> | Unshielded Twisted Pair (UTP) | 20 m         |

|                    |                               |      |
|--------------------|-------------------------------|------|
| <b>100BASE-T1</b>  | Shielded Twisted Pair (STP)   | 50 m |
| <b>1000BASE-T1</b> | Unshielded Twisted Pair (UTP) | 15 m |
| <b>1000BASE-T1</b> | Shielded Twisted Pair (STP)   | 40 m |

### 3 Connection Diagram



Note:

There must be one device set as Master, Another One set as Slave.  
Both of the devices should set to the same Speed.

### 4 User Guide for Linux Os

We use 2 Raspberry PI5, one as client(Master) and one as server(Slave)

Hardware connection as chapter 3.

#### 4.1 Install iperf3

Open terminal window of both raspberry pi5.

```
sudo apt-get install iperf3
```

**Note: Do not choose iperf3 as a daemon automatically.**

# BUELEC

**100/1000Base-T1-TX-E**

BUELEC 100/1000Base-T1 to RJ45  
(100/1000Base-TX) Automotive  
Ethernet Media Converter E Version

**Or it will fail when *you run next boot*.**

Download python scripts from our github.

```
sudo git clone https://github.com/buelec-tech/100-1000Base-T1-TX-L
```

[www.buelec-tech.com](http://www.buelec-tech.com)

[sales@buelec-tech.com](mailto:sales@buelec-tech.com)

[support@buelec-tech.com](mailto:support@buelec-tech.com)

## 4.2 Turn off WI-FI

## 4.3 Set IP And Ping

**Host A As Client (190.19.1.9), Host B As Server (190.19.1.90)**

| Host A (Raspberry PI5 A), Client                                                                                         | Host B (Raspberry PI5 B), Server                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <code>sudo ifconfig eth0 down</code><br><code>sudo ifconfig eth0 190.19.1.9</code><br><code>sudo ifconfig eth0 up</code> | <code>sudo ifconfig eth0 down</code><br><code>sudo ifconfig eth0 190.19.1.90</code><br><code>sudo ifconfig eth0 up</code> |

Open **Host A** Terminal window, run below command check if network connected.

```
sudo ping eth0 -i 190.19.1.90
```

- If work, then go ahead with chapter 4.4
- If it does not work, re-check connection follow chapter3

## 4.4 TCP Test

Open Host B (190.19.1.90) Terminal, Set as server

```
sudo iperf3 -s
```

**Note: if your system does not install iperf3, follow chapter 4.1**

Open Host A (190.19.1.9) Terminal, send data.

```
sudo iperf3 -c 190.19.1.90 -n 8000M -i 30
```

```
Connecting to host 190.19.1.90, port 5201
[ 5] local 190.19.1.9 port 48790 connected to 190.19.1.90 port 5201
[ ID] Interval           Transfer     Bitrate      Retr  Cwnd
[ 5]  0.00-30.00 sec   3.27 GBytes  937 Mbits/sec    0   518 KBytes
[ 5] 30.00-60.00 sec   3.27 GBytes  936 Mbits/sec    0   518 KBytes
[ 5] 60.00-71.63 sec   1.27 GBytes  937 Mbits/sec    0   518 KBytes
- - - - -
[ ID] Interval           Transfer     Bitrate      Retr
[ 5]  0.00-71.63 sec   7.81 GBytes  937 Mbits/sec    0      sender
[ 5]  0.00-71.64 sec   7.81 GBytes  937 Mbits/sec    0      receiver
```

## 4.4 UDP Test

Open Host B (190.19.1.90) Terminal, Set as server



```
sudo iperf3 -s
```

Open Host A (190.19.1.9) Terminal , Set as client and send data.

```
sudo iperf3 -c 190.19.1.90 -u -b 8000M -l 8k -n 1000M
```

```
Connecting to host 190.19.1.90, port 5201
[  5] local 190.19.1.9 port 38178 connected to 190.19.1.90 port 5201
[ ID] Interval           Transfer     Bitrate      Retr  Cwnd
[  5]  0.00-30.00 sec    3.27 GBytes  937 Mbits/sec    0   544 KBytes
[  5] 30.00-60.00 sec    3.27 GBytes  937 Mbits/sec    0   1.37 MBytes
[  5] 60.00-71.63 sec    1.27 GBytes  936 Mbits/sec    0   1.37 MBytes
-----
[ ID] Interval           Transfer     Bitrate      Retr
[  5]  0.00-71.63 sec    7.81 GBytes  937 Mbits/sec    0
[  5]  0.00-71.63 sec    7.81 GBytes  936 Mbits/sec    0
sender
receiver
iperf Done.
```

## 5 User Guide for Windows

We use 2 windows computers, one as client(Master) and one as server(Slave)

Hardware connection as chapter 3.

### 5.1 Install iperf3

Download from our github link:

<https://github.com/buelec-tech/100-1000Base-T1-TX-L> unzip iperf3.6\_64bit.zip

### 5.2 Turn off firewalls

### 5.3 Set IP And Ping

|                      |                       |
|----------------------|-----------------------|
| Computer A IP,Client | Computer B IP ,Server |
|----------------------|-----------------------|

# BUELEC

**100/1000Base-T1-TX-E**  
BUELEC 100/1000Base-T1 to RJ45  
(100/1000Base-TX) Automotive  
Ethernet Media Converter E Version

| General                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | General                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.</p> <p><input type="radio"/> Obtain an IP address automatically</p> <p><input checked="" type="radio"/> Use the following IP address:</p> <p>IP address: <input type="text" value="190 . 19 . 1 . 9"/></p> <p>Subnet mask: <input type="text" value="255 . 255 . 0 . 0"/></p> <p>Default gateway: <input type="text" value="190 . 19 . 1 . 1"/></p> <p><input type="radio"/> Obtain DNS server address automatically</p> <p><input checked="" type="radio"/> Use the following DNS server addresses:</p> <p>Preferred DNS server: <input type="text" value="8 . 8 . 8 . 8"/></p> <p>Alternate DNS server: <input type="text" value="8 . 8 . 4 . 4"/></p> <p><input type="checkbox"/> Validate settings upon exit <span>Advanced...</span></p> | <p>You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.</p> <p><input type="radio"/> Obtain an IP address automatically</p> <p><input checked="" type="radio"/> Use the following IP address:</p> <p>IP address: <input type="text" value="190 . 19 . 1 . 90"/></p> <p>Subnet mask: <input type="text" value="255 . 255 . 0 . 0"/></p> <p>Default gateway: <input type="text" value="190 . 19 . 1 . 1"/></p> <p><input type="radio"/> Obtain DNS server address automatically</p> <p><input checked="" type="radio"/> Use the following DNS server addresses:</p> <p>Preferred DNS server: <input type="text" value="8 . 8 . 8 . 8"/></p> <p>Alternate DNS server: <input type="text" value="8 . 8 . 4 . 4"/></p> <p><input type="checkbox"/> Validate settings upon exit <span>Advanced...</span></p> |
| 190.19.1.9<br>255.255.0.0<br>190.19.1.1<br>8.8.8.8<br>8.8.4.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 190.19.1.90<br>255.255.0.0<br>190.19.1.1<br>8.8.8.8<br>8.8.4.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Open Terminal (Admin) on Computer A, Switch to the iperf3 directory

Terminal

Terminal (Admin)

```
cd F:\iperf3.6_64bit
```

```
ping -i 190.19.1.9 190.19.1.90 # ping from 190.19.1.9(client)
```

```
PS F:\iperf3.6_64bit> ping -i 190.19.1.9 190.19.1.90
```

```
Pinging 190.19.1.90 with 32 bytes of data:
```

```
Reply from 190.19.1.90: bytes=32 time=2ms TTL=128
```

```
Reply from 190.19.1.90: bytes=32 time=2ms TTL=128
```

```
Reply from 190.19.1.90: bytes=32 time=2ms TTL=128
```

```
Reply from 190.19.1.90: bytes=32 time=3ms TTL=128
```

```
Ping statistics for 190.19.1.90:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
    Approximate round trip times in milli-seconds:
```

```
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

[www.buelec-tech.com](http://www.buelec-tech.com)

[sales@buelec-tech.com](mailto:sales@buelec-tech.com)

[support@buelec-tech.com](mailto:support@buelec-tech.com)

## 5.4 100M Test

Open Host B (190.19.1.90) Terminal, Set as server

```
.\iperf3.exe -B 190.19.1.90 -s # Host B listen
```

Open Host A (190.19.1.9) Terminal, Set as client and send data.

```
.\iperf3.exe -c 190.19.1.90 -B 190.19.1.9 -w 100M -t 10 # Host A send data
```

```
PS F:\iperf3.6_64bit> .\iperf3.exe -c 190.19.1.90 -B 190.19.1.9 -w 100M -t 10
warning: Ignoring nonsense TCP MSS 0
Connecting to host 190.19.1.90, port 5201
[ 5] local 190.19.1.9 port 9557 connected to 190.19.1.90 port 5201
[ ID] Interval           Transfer     Bitrate
[ 5] 0.00-1.00 sec      111 MBytes  933 Mbits/sec
[ 5] 1.00-2.00 sec      11.2 MBytes  94.4 Mbits/sec
[ 5] 2.00-3.00 sec      11.4 MBytes  95.3 Mbits/sec
[ 5] 3.00-4.00 sec      11.4 MBytes  95.5 Mbits/sec
[ 5] 4.00-5.00 sec      11.2 MBytes  94.4 Mbits/sec
[ 5] 5.00-6.00 sec      11.4 MBytes  95.3 Mbits/sec
[ 5] 6.00-7.00 sec      11.2 MBytes  94.4 Mbits/sec
[ 5] 7.00-8.00 sec      11.4 MBytes  95.5 Mbits/sec
[ 5] 8.00-9.00 sec      11.2 MBytes  94.4 Mbits/sec
[ 5] 9.00-10.00 sec     11.4 MBytes  95.4 Mbits/sec
-- -- -- -- --
[ ID] Interval           Transfer     Bitrate
[ 5] 0.00-10.00 sec     213 MBytes  179 Mbits/sec
[ 5] 0.00-10.13 sec     114 MBytes  94.6 Mbits/sec
sender
receiver
```

## 5.4 1000M Test

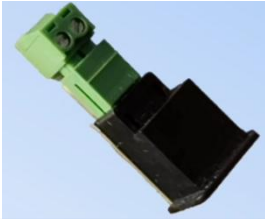
Open Host B (190.19.1.90) Terminal, Set as server

```
.\iperf3.exe -B 190.19.1.90 -s # Host B listen
```

Open Host A (190.19.1.9) Terminal, Set as client and send data.

```
.\iperf3.exe -c 190.19.1.90 -B 190.19.1.9 -w 100M -t 1
# -c <host address> ,-B <Client address >
```

## 6 Packing List

| No. | Item                                                                                                                                    | Quantity | Unit |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------|----------|------|
| 1   | 1000BASE-T1-TX-E Converter                                                                                                              | 1        | pcs  |
| 2   | 2-Pin Terminal Block (15EDG3.81mm-2P)                                                                                                   | 1        | pcs  |
| 3   | Type-C USB Power Cable<br>DC-Jack Power Adapter 12V                                                                                     | 1        | pcs  |
| 4   | CAT6 Ethernet Cable                                                                                                                     | 1        | pcs  |
| 5   | TE MATEnet (1x1) 150mm pigtail to terminal block<br>   | 1        | pcs  |
| 6   | Rosenberger HMTD 150mm pigtail to terminal block<br> | 1        | pcs  |

### 6.1 Accessories Recommend

<https://www.buelec-tech.com/product/1000base-t1-cable-h-mdt/>

<https://www.buelec-tech.com/product/1000base-t1-cable-matenet-connector/>

# BUELEC

**100/1000Base-T1-TX-E**

BUELEC 100/1000Base-T1 to RJ45  
(100/1000Base-TX) Automotive  
Ethernet Media Converter E Version

